

DAWSON, YT, Canada

WMO: 719660

Lat: 64.0446N

Lon: 139.1268W

Elev: 370

StdP: 96.96

Time Zone: -8.00 (NAP)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -47.6	(c) -44.9	(d) -52.8	(e) 0.0	(f) -47.5	(g) -49.7	(h) 0.0	(i) -44.8	(j) 4.9	(k) -13.1	(l) 4.2	(m) -14.9	(n) 0.1	(o) 60	(p) 0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	27.2	14.9	25.2	14.2	23.4	13.4	16.5	24.1	15.5	22.4	14.7	21.0	2.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.9	10.3	17.6	12.8	9.6	16.9	11.8	9.0	16.4	47.5	24.2	44.7	22.4	42.2	21.1	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.4	(b) 4.7	(c) 4.0	(e) -46.8	(f) 30.3	(g) 3.4	(h) 1.9	(i) -49.2	(j) 31.7	(k) -51.2	(l) 32.9	(m) -53.1	(n) 34.0	(o) -55.6	(p) 35.4		
(a) 5.4	(b) 4.7	(c) 4.0	(e) -46.7	(f) 18.4	(g) 3.4	(h) 1.0	(i) -49.1	(j) 19.2	(k) -51.1	(l) 19.8	(m) -53.0	(n) 20.4	(o) -55.4	(p) 21.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.8	-25.7	-20.5	-12.8	0.1	8.6	14.2	15.7	12.4	6.3	-3.9	-17.9	-22.7		
	DBStd	16.34	11.12	9.08	8.56	5.61	4.09	2.90	2.36	3.21	3.81	6.15	9.09	9.45		
	HDD10.0	5456	1106	854	707	298	77	3	1	12	119	430	836	1013		
	HDD18.3	8073	1364	1087	965	548	303	127	87	185	362	688	1086	1272		
	CDD10.0	432	0	0	0	0	33	129	176	87	7	0	0	0		
	CDD18.3	9	0	0	0	0	0	4	4	1	0	0	0	0		
	CDH23.3	407	0	0	0	0	32	144	160	71	0	0	0	0		
	CDH26.7	73	0	0	0	0	3	29	25	17	0	0	0	0		
	WSAvg	1.2	0.6	0.8	1.2	1.7	1.9	1.6	1.5	1.4	1.4	1.1	0.8	0.6		
	PrecAvg	324	17	15	11	10	24	37	48	41	35	34	26	23		
(15) Precipitation	PrecMax	438	41	36	41	25	57	52	95	83	72	72	55	47		
	PrecMin	246	3	3	0	0	6	9	17	9	6	13	10	6		
	PrecStd	61	10	11	11	8	14	13	21	19	20	17	13	12		
	PrecStd	61	10	11	11	8	14	13	21	19	20	17	13	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.1	2.5	8.3	17.8	26.6	30.0	29.3	28.9	19.6	11.7	0.6	0.0		
		MCWB	-0.4	-0.1	2.6	8.0	12.9	16.4	15.7	16.2	11.5	6.8	-0.4	-1.3		
	2%	DB	-4.9	-2.8	5.4	14.1	23.6	26.9	26.9	25.2	17.3	7.8	-2.6	-5.5		
		MCWB	-5.4	-3.7	1.2	6.0	11.8	14.5	15.0	15.1	10.2	4.2	-3.1	-6.0		
	5%	DB	-8.9	-6.1	2.5	11.8	21.1	24.8	25.1	22.3	15.6	5.1	-4.8	-8.1		
		MCWB	-9.4	-6.7	-0.6	4.5	10.5	13.5	14.7	13.9	9.3	2.7	-5.4	-8.5		
	10%	DB	-11.8	-9.0	-0.2	9.7	18.3	22.9	23.3	19.8	13.7	2.9	-6.8	-10.7		
		MCWB	-12.2	-9.6	-2.3	3.5	9.2	12.6	14.1	12.8	8.5	1.5	-7.3	-11.0		
	0.4%	WB	-0.5	0.1	3.4	8.5	14.2	17.9	17.6	17.5	12.4	7.6	-0.2	-1.2		
		MCDB	0.2	2.4	6.9	17.4	23.4	26.3	25.5	25.7	17.3	10.4	0.4	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-5.4	-3.6	1.5	6.2	12.5	16.0	16.5	16.1	11.1	4.7	-3.1	-5.9		
		MCDB	-4.8	-2.6	4.9	13.1	21.8	23.9	23.8	23.5	15.6	7.0	-2.5	-5.6		
	5%	WB	-9.3	-6.8	-0.3	4.9	11.2	14.7	15.7	14.7	10.1	3.0	-5.3	-8.5		
		MCDB	-8.9	-6.1	2.4	10.8	19.4	21.9	22.5	20.4	14.4	4.8	-4.8	-8.2		
	10%	WB	-12.2	-9.6	-2.3	3.7	9.8	13.7	15.0	13.6	9.1	1.5	-7.3	-11.0		
		MCDB	-11.8	-9.1	-0.1	9.1	17.3	20.7	21.4	18.5	12.8	2.8	-6.8	-10.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	10.6	16.4	14.6	14.6	14.7	13.6	13.2	12.1	8.2	7.4	7.4		
		MCDBR	9.5	12.7	16.1	16.5	19.2	19.1	18.1	17.7	15.4	10.3	7.7	9.3		
		MCWBR	9.2	11.7	12.3	9.4	9.7	9.1	8.9	9.3	9.2	7.3	7.3	8.9		
		MCDBR	9.7	12.7	14.9	15.2	17.2	16.3	15.1	15.1	13.4	9.1	7.5	9.4		
	5% WB	MCWBR	9.4	11.7	11.5	8.9	9.1	8.2	7.8	8.4	8.4	6.7	7.1	9.1		
		MCWBR	9.4	11.7	11.5	8.9	9.1	8.2	7.8	8.4	8.4	6.7	7.1	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.201	0.244	0.260	0.349	0.368	0.385	0.399	0.380	0.316	0.262	0.181	0.175		
	taud		1.939	2.219	2.266	2.079	2.317	2.310	2.294	2.353	2.517	2.456	1.973	1.923		
	Ebn at Noon		368	697	846	813	840	830	803	784	784	688	400	175		
	Edh at Noon		28	55	81	125	109	113	112	96	66	44	25	11		
(44) All-Sky Solar Radiation	RadAvg		0.20	0.98	2.77	4.84	5.51	5.60	4.83	3.84	2.40	0.99	0.29	0.07		
	RadStd		0.01	0.06	0.14	0.21	0.31	0.41	0.36	0.34	0.20	0.13	0.03	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	+2.75	+3.17	N/A	N/A	N/A	N/A	N/A	+55	N/A
	N/A	+2.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page